

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (10 points) Let $A = \begin{bmatrix} 1 & 3 & 1 & 1 \\ 2 & 6 & 4 & 3 \\ -2 & -6 & 0 & -1 \end{bmatrix}$.

(a) Transform A into the reduced echelon form R .

$$\begin{aligned}
 A = \begin{bmatrix} 1 & 3 & 1 & 1 \\ 2 & 6 & 4 & 3 \\ -2 & -6 & 0 & -1 \end{bmatrix} &\rightarrow \begin{bmatrix} 1 & 3 & 1 & 1 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 2 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 1 & 1 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \\
 &\rightarrow \begin{bmatrix} 1 & 3 & 1 & 1 \\ 0 & 0 & 1 & 1/2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \\
 &\rightarrow \begin{bmatrix} 1 & 3 & 0 & 1/2 \\ 0 & 0 & 1 & 1/2 \\ 0 & 0 & 0 & 0 \end{bmatrix} = R. \\
 &\Rightarrow \text{rank}(A) = 2
 \end{aligned}$$

(b) Find the dimension and a basis for the column space of A , $C(A)$.

$$\dim(C(A)) = 2 = \text{rank}(A)$$

$$\text{Basis: } \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix}, \begin{bmatrix} 1 \\ 4 \\ 0 \end{bmatrix}$$

(c) Find the dimension and a basis for the row space of A , $C(A^T)$.

$$\dim C(A^T) = 2 = \text{rank}(A)$$

$$\text{Basis: } \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 6 \\ 4 \end{bmatrix}$$

$$\left(\text{or } \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1/2 \end{bmatrix} \right)$$

(d) Find the dimension and a basis for the nullspace of A , $N(A)$.

$$\dim(N(A)) = 4 - \text{rank}(A) = 4 - 2 = 2$$

Solve Special solution $Rx = 0$

i) Set $x_2 = 1, x_4 = 0$, we get $x_1 = -3, x_3 = 0$.

$$s_1 = (-3, 1, 0, 0)$$

ii) Set $x_2 = 0, x_4 = 1$, we get $x_1 = -1/2, x_3 = -1/2$.

$$s_2 = (-1/2, 0, -1/2, 1)$$

Thus Basis of $N(A)$

$$\begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1/2 \\ 0 \\ -1/2 \\ 1 \end{bmatrix}$$

(e) What is the dimension of the left nullspace of A , $N(A^T)$?

$$\dim(N(A^T)) = 3 - \text{rank}(A) = 3 - 2 = 1$$